The optimal solution to the first problem is $\bar{x}_1 = 1.39$ and the optimal objective value is $\bar{z}_1 = -1.93$. The optimal solution to the second problem is $\bar{x}_2 = 2.39$ and the optimal objective value is $\bar{z}_2 = -5.71$. Thus, minimum $\{\bar{z}_1 + \hat{v}_1, \bar{z}_2 + \hat{v}_2\} = \bar{z}_1 + \hat{v}_1 = \bar{z}_2 + \hat{v}_2 = -1.93$. Therefore, we can introduce either the grid point $\bar{x}_1 = 1.39$ or the grid point $\bar{x}_2 = 2.39$. Note that

$$\sum_{j=1}^{2} \bar{z}_j - \sum_{i=1}^{2} \hat{u}_i p_i = -23.74 \quad \text{and} \quad f(\hat{\mathbf{x}}) = -21.88.$$

By part 1 of Theorem 11.3.5, the optimal objective value of the original problem lies between $-23.74$ and $-21.88$. Thus, if we stop the algorithm at this stage, we would have a feasible solution $\hat{\mathbf{x}} = (1.25, 3.75, 0)'$ whose objective value is $-21.88$. and we would also know that a lower bound on the optimal objective value to the original problem is $-23.74$. If more accuracy is desired, the process would continue by introducing the new grid point $x_{31} = 1.39$ or the new grid point $x_{32} = 2.39$.

## 11.4 Linear Fractional Programming

In this section, we consider a problem in which the objective function is the ratio of two linear functions and the constraints are linear. Such problems are called *linear fractional programming problems* and can be stated precisely as follows:

$$\begin{aligned} \text{Minimize} \quad & \frac{\mathbf{p}'\mathbf{x} + \alpha}{\mathbf{q}'\mathbf{x} + \beta} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

where $\mathbf{p}$ and $\mathbf{q}$ are $n$ vectors, $\mathbf{b}$ is an $m$ vector, $\mathbf{A}$ is an $m \times n$ matrix, and $\alpha$ and $\beta$ are scalars. As we shall soon observe, if an optimal solution for a linear fractional program exists, then an extreme point optimum exists. Furthermore, every local minimum is a global minimum. Hence, a procedure that moves from one extreme point to an adjacent one seems very attractive for solving such a problem. Lemma 11.4.1 below gives some important properties of the objective function.

### 11.4.1 Lemma

Let $f(\mathbf{x}) = (\mathbf{p}'\mathbf{x} + \alpha)/(\mathbf{q}'\mathbf{x} + \beta)$, and let $S$ be a convex set such that $\mathbf{q}'\mathbf{x} + \beta \neq 0$ over $S$. Then, $f$ is both pseudoconvex and pseudoconcave over $S$.

*Proof*

First, note that either $\mathbf{q}'\mathbf{x} + \beta > 0$ for all $\mathbf{x} \in S$ or $\mathbf{q}'\mathbf{x} + \beta < 0$ for all $\mathbf{x} \in S$. Otherwise, there exist an $\mathbf{x}_1$ and $\mathbf{x}_2$ in $S$ such that $\mathbf{q}'\mathbf{x}_1 + \beta > 0$ and $\mathbf{q}'\mathbf{x}_2 + \beta < 0$; and, hence, for some convex combination $\mathbf{x}$ of $\mathbf{x}_1$ and $\mathbf{x}_2$, $\mathbf{q}'\mathbf{x} + \beta = 0$, contradicting our assumption. We first show that $f$ is pseudoconvex. Suppose that $\mathbf{x}_1, \mathbf{x}_2 \in S$ with $(\mathbf{x}_2 - \mathbf{x}_1)'\nabla f(\mathbf{x}_1) \geq 0$. We need to show that $f(\mathbf{x}_2) \geq f(\mathbf{x}_1)$. Note that

$$\nabla f(\mathbf{x}_1) = \frac{(\mathbf{q}'\mathbf{x}_1 + \beta)\mathbf{p} - (\mathbf{p}'\mathbf{x}_1 + \alpha)\mathbf{q}}{(\mathbf{q}'\mathbf{x}_1 + \beta)^2}$$

Since $(\mathbf{x}_2 - \mathbf{x}_1)'\nabla f(\mathbf{x}_1) \geq 0$ and since $(\mathbf{q}'\mathbf{x}_1 + \beta)^2 > 0$, it follows that

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Th. 3.5.11 Let S be a nonempty open convex set in $E_n$ and let $f: S \to E_1$ be a differentiable pseudoconvex fn on S. Then f is both strictly quasiconvex and quasiconcave (Ch-3)



$$0 \le (\mathbf{x}_2 - \mathbf{x}_1)'[(\mathbf{q}'\mathbf{x}_1 + \beta)\mathbf{p} - (\mathbf{p}'\mathbf{x}_1 + \alpha)\mathbf{q}]$$
$$= (\mathbf{p}'\mathbf{x}_2 + \alpha)(\mathbf{q}'\mathbf{x}_1 + \beta) - (\mathbf{q}'\mathbf{x}_2 + \beta)(\mathbf{p}'\mathbf{x}_1 + \alpha)$$

Therefore, $(\mathbf{p}'\mathbf{x}_2 + \alpha)(\mathbf{q}'\mathbf{x}_1 + \beta) \ge (\mathbf{q}'\mathbf{x}_2 + \beta)(\mathbf{p}'\mathbf{x}_1 + \alpha)$. But since $\mathbf{q}'\mathbf{x}_1 + \beta$ and $\mathbf{q}'\mathbf{x}_2 + \beta$ are both either positive or negative, dividing by $(\mathbf{q}'\mathbf{x}_1 + \beta)(\mathbf{q}'\mathbf{x}_2 + \beta) > 0$, we get

$$\frac{\mathbf{p}'\mathbf{x}_2 + \alpha}{\mathbf{q}'\mathbf{x}_2 + \beta} \ge \frac{\mathbf{p}'\mathbf{x}_1 + \alpha}{\mathbf{q}'\mathbf{x}_1 + \beta}, \text{ that is, } f(\mathbf{x}_2) \ge f(\mathbf{x}_1)$$

Therefore, $f$ is pseudoconvex. Similarly, it can be shown that $(\mathbf{x}_2 - \mathbf{x}_1)'\nabla f(\mathbf{x}_1) \le 0$ implies that $f(\mathbf{x}_2) \le f(\mathbf{x}_1)$, and, hence, $f$ is pseudoconcave, and the proof is complete.

Several implications of the above lemma for a linear fractional programming problem may be noted.

1. Since the objective function is both pseudoconvex and pseudoconcave over $S$, then by Theorem 3.5.11, it is also quasiconvex, quasiconcave, strictly quasiconvex, and strictly quasiconcave.
2. Since the objective function is both pseudoconvex and pseudoconcave, then, by Theorem 4.3.8, a point satisfying the KKT conditions for a minimization problem is also a global minimum over the feasible region. Likewise, a point satisfying the KKT conditions for a maximization problem is also a global maximum over the feasible region.
3. Since the objective function is strictly quasiconvex and strictly quasiconcave, then, by Theorem 3.5.6, a local minimum is also a global minimum over the feasible region. Likewise, a local maximum is also a global maximum over the feasible region.
4. Since the objective function is quasiconcave and quasiconvex, if the feasible region is bounded, then, by Theorem 3.5.3, the objective function has a minimum at an extreme point of the feasible region and also has a maximum at an extreme point of the feasible region.

The above facts about the objective function $f$ give very useful results that can be used to develop suitable computational procedures for solving the fractional programming problem. In particular, we may search among the extreme points of the polyhedral set $\{\mathbf{x}: \mathbf{A}\mathbf{x} = \mathbf{b}, \mathbf{x} \ge \mathbf{0}\}$ until a KKT point is reached. We now show that the convex–simplex method gives a convenient solution procedure.

## Minimization by the Convex–Simplex Method

Because of the special structure of the objective function $f$, the convex–simplex method simplifies into a minor modification of the simplex method of linear programming. Suppose that we are given an extreme point of the feasible region with basis $\mathbf{B}$ such that $\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} > \mathbf{0}$ and $\mathbf{x}_N = \mathbf{0}$. Recall from Section 10.7 that the convex–simplex method increases or decreases one of the nonbasic variables and then modifies the basic variables accordingly. Since the current point is an extreme point with $\mathbf{x}_N = \mathbf{0}$, decreasing a nonbasic variable is not permitted, as it would violate the nonnegativity restriction. Thus, the direction-finding process simplifies as follows. Let $\mathbf{r}_N$ denote the nonbasic components of the reduced gradient vector $\mathbf{r}' = \nabla f(\mathbf{x})' - \nabla_B f(\mathbf{x})'\mathbf{B}^{-1}\mathbf{A}$, so that

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$$\mathbf{r}_N' = \nabla_N f(\mathbf{x})' + \nabla_B f(\mathbf{x})' \mathbf{B}^{-1}\mathbf{N}$$

By Theorem 10.5.1, if $\mathbf{r}_N \geq \mathbf{0}$, then the current point is a KKT point, and we must stop. Otherwise, let $-r_j = \text{maximum } \{-r_i : r_i \leq 0\}$, where $r_i$ is the $i$th component of $\mathbf{r}_N$. The nonbasic variable $x_j$ is increased, and the basic variables are modified to maintain feasibility. This is equivalent to moving along the direction $\mathbf{d}$ whose nonbasic and basic components $\mathbf{d}_N$ and $\mathbf{d}_B$ are given as follows. The direction $\mathbf{d}_N$ is a vector of zeros, except for a 1 at the $j$th position, and $\mathbf{d}_B = -\mathbf{B}^{-1}\mathbf{a}_j$, where $\mathbf{a}_j$ is the $j$th column of $\mathbf{A}$. By Theorem 10.6.1, $\mathbf{d}$ is an improving feasible direction. As we shall see by Lemma 11.4.2, no line search along the direction $\mathbf{d}$ is needed. Indeed, due to the special structure of the objective function, if $\nabla f(\mathbf{x})'\mathbf{d} < 0$, then the function $f$ continues to decrease by moving along $\mathbf{d}$. Thus, we must move along $\mathbf{d}$ as far as possible. Since moving along $\mathbf{d}$ is equivalent to increasing a nonbasic variable and adjusting the basic variables, we move along $\mathbf{d}$ until a basic variable drops to zero and leaves the basis, producing an adjacent extreme point. The whole process is then repeated.

### 11.4.2 Lemma

Let $f(\mathbf{x}) = (\mathbf{p}'\mathbf{x} + \alpha)/(\mathbf{q}'\mathbf{x} + \beta)$, and let $S$ be a convex set. Furthermore, suppose that $\mathbf{q}'\mathbf{x} + \beta \neq 0$ on $S$. Given $\mathbf{x} \in S$, let $\mathbf{d}$ be such that $\nabla f(\mathbf{x})'\mathbf{d} < 0$. Then, $f(\mathbf{x} + \lambda\mathbf{d})$ is a decreasing function of $\lambda$.

*Proof*

Note that

$$\nabla f(\mathbf{y}) = \frac{(\mathbf{q}'\mathbf{y} + \beta)\mathbf{p} - (\mathbf{p}'\mathbf{y} + \alpha)\mathbf{q}}{(\mathbf{q}'\mathbf{y} + \beta)^2} \tag{11.47}$$

Letting $\mathbf{y} = \mathbf{x} + \lambda\mathbf{d}$, $s = [\mathbf{q}'(\mathbf{x} + \lambda\mathbf{d}) + \beta]^2 > 0$, and $s' = (\mathbf{q}'\mathbf{x} + \beta)^2 > 0$, we get

$$\nabla f(\mathbf{x} + \lambda\mathbf{d}) = \frac{[\mathbf{q}'(\mathbf{x} + \lambda\mathbf{d}) + \beta]\mathbf{p} - [\mathbf{p}'(\mathbf{x} + \lambda\mathbf{d}) + \alpha]\mathbf{q}}{s}$$

$$= \frac{s'}{s}\nabla f(\mathbf{x}) + \frac{\lambda}{s}[(\mathbf{q}'\mathbf{d})\mathbf{p} - (\mathbf{p}'\mathbf{d})\mathbf{q}]$$

Therefore,

$$\nabla f(\mathbf{x} + \lambda\mathbf{d})'\mathbf{d} = \frac{s'}{s}\nabla f(\mathbf{x})'\mathbf{d} + \frac{\lambda}{s}[(\mathbf{q}'\mathbf{d})(\mathbf{p}'\mathbf{d}) - (\mathbf{p}'\mathbf{d})(\mathbf{q}'\mathbf{d})]$$

$$= \frac{s'}{s}\nabla f(\mathbf{x})'\mathbf{d} \tag{11.48}$$

Now let $\theta(\lambda) = f(\mathbf{x} + \lambda\mathbf{d})$. Then, by (11.48), $\theta'(\lambda) = \nabla f(\mathbf{x} + \lambda\mathbf{d})'\mathbf{d} < 0$ for all $\lambda$, and the result follows.

To summarize, given the extreme point $\mathbf{x}$ and the direction $\mathbf{d}$ with $\nabla f(\mathbf{x})'\mathbf{d} < 0$, no minimization of $f$ along $\mathbf{d}$ is necessary, since $f(\mathbf{x} + \lambda\mathbf{d})$ is a decreasing function of $\lambda$. Therefore, we move along $\mathbf{d}$ as much as possible, that is, until an adjacent extreme point is reached, and we then repeat the process. A precise summary of the algorithm utilizing a tableau format for updating the extreme points generated is presented below.

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### ...ary of the Fractional Programming Algorithm of Gilmore and Gomory

We present below a method credited to Gilmore and Gomory [1963] for solving a linear fractional program of the form to minimize $(\mathbf{p}'\mathbf{x} + \alpha)/(\mathbf{q}'\mathbf{x} + \beta)$ subject to $\mathbf{x} \in S = \{\mathbf{x}: \mathbf{A}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$. We will assume that the set $S$ is bounded and that $\mathbf{q}'\mathbf{x} + \beta \neq 0$ for all $\mathbf{x} \in S$.

*Initialization Step* Find a starting basic feasible solution $\mathbf{x}_1$ to the system $\mathbf{A}\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$. Form the corresponding tableau represented by $\mathbf{x}_B + \mathbf{B}^{-1}\mathbf{N}\mathbf{x}_N = \mathbf{B}^{-1}\mathbf{b}$. Let $k = 1$ and go to the main step.

*Main Step*

1. Compute the vector $\mathbf{r}_N' = \nabla_N f(\mathbf{x}_k)' - \nabla_B f(\mathbf{x}_k)'\mathbf{B}^{-1}\mathbf{N}$. If $\mathbf{r}_N \geq \mathbf{0}$, stop; the current point $\mathbf{x}_k$ is an optimal solution. Otherwise, go to step 2.
2. Let $-r_j = \text{maximum } \{-r_i : r_i \leq 0\}$, where $r_i$ is the $i$th component of $\mathbf{r}_N$: Determine the basic variable $\mathbf{x}_B$, to leave the basis by the following minimum ratio test:

$$\frac{\bar{b}_r}{y_{rj}} = \underset{1 \leq i \leq m}{\text{minimum}} \left\{ \frac{\bar{b}_i}{y_{ij}} : y_{ij} > 0 \right\}$$

where $\bar{\mathbf{b}} = \mathbf{B}^{-1}\mathbf{b}$, $\mathbf{y}_j = \mathbf{B}^{-1}\mathbf{a}_j$, and $\mathbf{a}_j$ is the $j$th column of $\mathbf{A}$. Go to step 3.
3. Replace the variable $x_{B_r}$ by the variable $x_j$. Update the tableau correspondingly by pivoting at $y_{rj}$. Let the current solution be $\mathbf{x}_{k+1}$. Replace $k$ by $k + 1$ and go to step 1.

Exercise 11.38 shows that the reduced gradient $\mathbf{r}_N$ could be readily computed if two additional rows, one corresponding to $\mathbf{p}'\mathbf{x} + \alpha$ and the other corresponding to $\mathbf{q}'\mathbf{x} + \beta$ are introduced and carried forward at each iteration.

### ...ite Convergence

We now assume that $\mathbf{x}_B > \mathbf{0}$ for each extreme point. The algorithm moves from one extreme point to another. By Lemma 11.4.2 and the nondegeneracy assumption, the objective function strictly decreases at each iteration so that the extreme points generated are distinct. There is only a finite number of these points, and, hence, the algorithm stops in a finite number of steps. At termination, the reduced gradient is nonnegative resulting in a KKT point; and, by Lemma 11.4.1, this point is indeed an optimal point.

### 3 Example

Consider the following problem:

$$\begin{aligned} \text{Minimize} \quad & \frac{-2x_1 + x_2 + 2}{x_1 + 3x_2 + 4} \\ \text{subject to} \quad & -x_1 + x_2 \leq 4 \\ & x_2 \leq 6 \\ & 2x_1 + x_2 \leq 14 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Figure 11.5 shows the feasible region with the extreme points (0, 0), (0, 4), (2, 6), (4, 6), and (7, 0). The objective at these points is 0.5, 0.375, 0.167, 0.0, and −1.09, respectively, and, hence, the optimal point is (7, 0).

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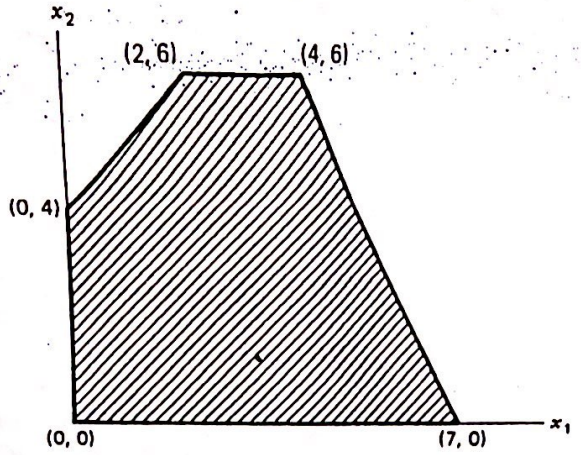



**Figure 11.5** The feasible region for Example 11.4.3.

Introducing the slack variables $x_3$, $x_4$, and $x_5$, we get the initial extreme point $\mathbf{x}_1 = (0, 0, 4, 6, 14)'$.

### *Iteration 1*

The tableau below summarizes the computations for this iteration.

| | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | RHS |
|---|---|---|---|---|---|---|
| $\nabla f(\mathbf{x}_1)$ | $-\frac{10}{16}$ | $-\frac{2}{16}$ | 0 | 0 | 0 | — |
| $x_3$ | −1 | 1 | 1 | 0 | 0 | 4 |
| $x_4$ | 0 | 1 | 0 | 1 | 0 | 6 |
| $x_5$ | 2 | 1 | 0 | 0 | 1 | 14 |
| $\mathbf{r}$ | $-\frac{10}{16}$ | $-\frac{2}{16}$ | 0 | 0 | 0 | — |

We have $\mathbf{q}'\mathbf{x}_1 + \beta = 4$ and $\mathbf{p}'\mathbf{x}_1 + \alpha = 2$. Hence, from (11.47), we get $\nabla f(\mathbf{x})' = (-\frac{10}{16}, -\frac{2}{16}, 0, 0, 0)$. $\nabla_N f(\mathbf{x})' = (-\frac{10}{16}, -\frac{2}{16})$ and $\nabla_B f(\mathbf{x})' = (0, 0, 0)$. The columns of $x_1$ and $x_2$ give $\mathbf{B}^{-1}\mathbf{N}$, and we get

$$\mathbf{r}_N' = (r_1, r_2) = \nabla_N f(\mathbf{x}_1)' - \nabla_B f(\mathbf{x}_1)'\mathbf{B}^{-1}\mathbf{N}$$

$$= (-\tfrac{10}{16}, -\tfrac{2}{16}) - (0, 0, 0)\begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 2 & 1 \end{bmatrix} = (-\tfrac{10}{16}, -\tfrac{2}{16})$$

Note that $\mathbf{r}_N' = (r_3, r_4, r_5) = (0, 0, 0)$. Since maximum $\{-r_1, -r_2, -r_3, -r_4, -r_5\} = \frac{10}{16}$, $x_1$ enters the basis. By the minimum ratio test, $x_5$ leaves the basis.

### *Iteration 2*

The computations for this iteration are summarized below.

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11.31 In Section 11.3, we approximated a separable programming problem using the λ-form. An alternative, called the δ-*form*, was considered in Exercise 11.29. Consider a variable $x$ in the interval $[a, b]$ and grid points $\mu_1 = a, \mu_2, \ldots, \mu_k = b$. Then, using the λ- and δ-forms, $x$ can be represented as

1: $x = \sum_{j=1}^{k} \lambda_j \mu_j \qquad \sum_{j=1}^{k} \lambda_j = 1 \qquad \lambda_j \geq 0 \qquad$ for $j = 1, \ldots, k$

where $\lambda_p \lambda_q = 0$ if $\mu_p$ and $\mu_q$ are not adjacent.

2. $x = \mu_1 + \sum_{j=1}^{k-1} \Delta_j \delta_j \qquad 0 \leq \delta_j \leq 1 \qquad$ for $j = 1, \ldots, k-1$

$\delta_i > 0 \Rightarrow \delta_j = 1 \qquad$ for $j < i$

Show that the two forms are related by the relationship

$$\lambda_j = \begin{cases} \delta_{j-1} - \delta_j & \text{if } j = 1, \ldots, k-1 \\ \delta_{j-1} & \text{if } j = k \end{cases}$$

where $\delta_0 = 1$. In particular, show that this relationship could be written in vector form as $\boldsymbol{\lambda} = \mathbf{T}\boldsymbol{\delta}$, where $\mathbf{T}$ is an upper triangular matrix.

11.32 Solve the following problem by the two linear fractional programming algorithms discussed in Section 11.4:

$$\text{Minimize} \quad \frac{-2x_1 + 3x_2 + 5x_3 + 2}{x_1 + 2x_2 + x_3 + 2}$$

$$\text{subject to} \quad \begin{aligned} 2x_1 + 3x_2 + x_3 &\leq 12 \\ x_1 + 2x_2 &\geq 2 \\ x_1 + x_3 &\leq 8 \\ x_1, \quad x_2, \quad x_3 &\geq 0 \end{aligned}$$

11.33 Consider the following problem:

$$\text{Maximize} \quad \frac{8x_1 + 6x_2 - 5}{-4x_1 + 2x_2 - 40}$$

$$\text{subject to} \quad \begin{aligned} x_1 + x_2 &\leq 10 \\ 3x_1 - 5x_2 &\leq 6 \\ x_1, x_2 &\geq 0 \end{aligned}$$

a. Solve the problem by the method of Gilmore and Gomory.

b. Solve the problem by the method of Charnes and Cooper.

11.34 Suppose that the region $\{\mathbf{x} : \mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$ is unbounded. Further, suppose that an improving feasible direction $\mathbf{d}$ is found while minimizing a linear fractional function over the above region. In particular, suppose that $\mathbf{d}_N$ consists of a vector of zeros, except for a 1 at position $j$ and $\mathbf{d}_B = -\mathbf{B}^{-1}\mathbf{a}_j \geq \mathbf{0}$. Is it necessarily true that the optimal objective is unbounded by moving from the current extreme point in the direction $\mathbf{d}$? If not, what are the possible cases that can be encountered?

11.35 Consider the function $f$ defined by

$$f(\mathbf{x}) = \frac{x_1 + 2x_2 - 6}{3x_1 - x_2 + 2}$$

a. Sketch the following sets in the $(x_1, x_2)$ plane, and determine whether they are convex:

$S = \{(x_1, x_2) : f(\mathbf{x}) \leq 2\}$

$S_1 = S \cap \{(x_1, x_2) : 3x_1 - x_2 + 2 > 0\}$

$S_2 = S \cap \{(x_1, x_2) : 3x_1 - x_2 + 2 < 0\}$

b. Is your conclusion in part a inconsistent with the fact that $f$ is quasiconvex on the region $\{(x_1, x_2) : 3x_1 - x_2 + 2 \neq 0\}$? Discuss.

11.36 Let

$$f(\mathbf{x}) = \frac{\mathbf{p}'\mathbf{x} + \alpha}{\mathbf{q}'\mathbf{x} + \beta}$$

and let $S = \{\mathbf{x} : \mathbf{q}'\mathbf{x} + \beta > 0\}$. Show directly that $f$ is quasiconvex, quasiconcave, strictly quasiconvex, and strictly quasiconcave on $S$.

11.37 Let $f : E_n \to E_1$ be quasiconcave, and let $\theta(\lambda) = f(\mathbf{x} + \lambda\mathbf{d})$, where $\mathbf{x}$ is a given vector and $\mathbf{d}$ is a given direction.

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| | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | RHS |
|---|---|---|---|---|---|---|
| $\nabla f(\mathbf{x}_2)$ | $-\frac{10}{121}$ | $\frac{47}{121}$ | 0 | 0 | 0 | — |
| $x_3$ | 0 | $\frac{3}{2}$ | 1 | 0 | $\frac{1}{2}$ | 11 |
| $x_4$ | 0 | 1 | 0 | 1 | 0 | 6 |
| $x_1$ | 1 | $\frac{1}{2}$ | 0 | 0 | $\frac{1}{2}$ | 7 |
| $\mathbf{r}$ | 0 | $\frac{52}{121}$ | 0 | 0 | $\frac{5}{121}$ | — |

When $x_1$ replaces $x_5$ in the basis, we get the point $\mathbf{x}_2' = (7, 0, 11, 6, 0)$. Now $\mathbf{q}'\mathbf{x}_2 + \beta = 11$ and $\mathbf{p}'\mathbf{x}_2 + \alpha = -12$, so that, from (11.47), we get $\nabla f(\mathbf{x}_2)' = (-\frac{10}{121}, \frac{47}{121}, 0, 0, 0)$. $\mathbf{B}^{-1}\mathbf{N}$ is given by the columns of $x_2$ and $x_5$ in the tableau, and we then get

$$\mathbf{r}_N' = (r_2, r_5) = \nabla_N f(\mathbf{x}_2)' - \nabla_B f(\mathbf{x}_2)'\mathbf{B}^{-1}\mathbf{N}$$

$$= (\tfrac{47}{121}, 0) - (0, 0, -\tfrac{10}{121}) \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$= (\tfrac{52}{121}, \tfrac{5}{121})$$

Since $\mathbf{r}_N \geq 0$, we stop with the optimal solution $x_1 = 7$ and $x_2 = 0$. The corresponding objective function value is $-1.09$.

### The Method of Charnes and Cooper [1962]

We now describe another procedure using the simplex method for solving a linear fractional programming problem. Consider the following problem:

$$\text{(LFP)} \quad \text{Minimize} \quad \frac{\mathbf{p}'\mathbf{x} + \alpha}{\mathbf{q}'\mathbf{x} + \beta}$$
$$\text{subject to} \quad \mathbf{A}\mathbf{x} \leq \mathbf{b}$$
$$\mathbf{x} \geq \mathbf{0}$$

Suppose that the set $S = \{\mathbf{x}: \mathbf{A}\mathbf{x} \leq \mathbf{b} \text{ and } \mathbf{x} \geq \mathbf{0}\}$ is compact, and suppose that $\mathbf{q}'\mathbf{x} + \beta > 0$ for each $\mathbf{x} \in S$. Letting $z = 1/(\mathbf{q}'\mathbf{x} + \beta)$ and $\mathbf{y} = z\mathbf{x}$, the above problem leads to the following linear program:

$$\text{(LP)} \quad \text{Minimize} \quad \mathbf{p}'\mathbf{y} + \alpha z$$
$$\text{subject to} \quad \mathbf{A}\mathbf{y} - \mathbf{b}z \leq \mathbf{0}$$
$$\mathbf{q}'\mathbf{y} + \beta z = 1$$
$$\mathbf{y} \geq \mathbf{0}$$
$$z \geq 0$$

First, note that, if $(\mathbf{y}, z)$ is a feasible solution to the above problem, then $z \geq 0$. This follows since, if $z = 0$, then $\mathbf{y} \neq \mathbf{0}$ must be such that $\mathbf{A}\mathbf{y} \leq \mathbf{0}$ and $\mathbf{y} \geq \mathbf{0}$, which means that $\mathbf{y}$ is a direction of $S$, violating the compactness assumption. We now demonstrate that, if $(\bar{\mathbf{y}}, \bar{z})$ is an optimal solution to the above linear program, then $\bar{\mathbf{x}} = \bar{\mathbf{y}}/\bar{z}$ is an optimal solution to the fractional program.

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Note that $A\bar{x} \leq b$ and $\bar{x} \geq 0$, so that $\bar{x}$ is a feasible solution to the fractional program. To show optimality of $\bar{x}$, let $x$ be such that $Ax \leq b$ and $x \geq 0$. Note that $q'x + \beta > 0$ by assumption, and that the vector $(y, z)$ is a feasible solution to the linear program, where $y = x/(q'x + \beta)$ and $z = 1/(q'x + \beta)$. Since $(\bar{y}, \bar{z})$ is an optimal solution to the linear program, $p'\bar{y} + \alpha\bar{z} \leq p'y + \alpha z$. Substituting for $\bar{y}$, $y$, and $z$, this inequality gives $\bar{z}(p'\bar{x} + \alpha) \leq (p'x + \alpha)/(q'x + \beta)$. The result immediately follows by dividing the left-hand side by $1 = q'\bar{y} + \beta\bar{z}$.

Now, if $q'x + \beta \leq 0$ for all $x \in S$, then letting $-z = 1/(q'x + \beta)$ and $y = zx$ gives the following linear program:

$$\begin{aligned} \text{Minimize} \quad & -p'y - \alpha z \\ \text{subject to} \quad & Ay - bz \leq 0 \\ & -q'y - \beta z = 1 \\ & y \geq 0 \\ & z \geq 0 \end{aligned}$$

In a fashion similar to that above, if $(\bar{y}, \bar{z})$ solves the above linear program, then $\bar{x} = \bar{y}/\bar{z}$ solves the fractional programming problem.

To summarize, we have shown that a fractional linear program could be solved by a linear programming problem with one additional variable and one additional constraint. The form of the linear program used depends on whether $q'x + \beta > 0$ for all $x \in S$ or $q'x + \beta < 0$ for all $x \in S$. If there exist $x_1, x_2 \in S$ such that $q'x_1 + \beta > 0$ and $q'x_2 + \beta < 0$, then the optimal solution to the fractional program is unbounded.

### 11.4.4 Example

Consider the following problem:

$$\begin{aligned} \text{Minimize} \quad & \frac{-2x_1 + x_2 + 2}{x_1 + 3x_2 + 4} \\ \text{subject to} \quad & -x_1 + x_2 \leq 4 \\ & 2x_1 + x_2 \leq 14 \\ & x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$$

The feasible region for this problem is shown in Figure 11.5. We solve this problem using the method of Charnes and Cooper. Note that the point (0, 0) is feasible and that at this point $x_1 + 3x_2 + 4 > 0$. Hence, the denominator is positive over the entire feasible region. The equivalent linear program is given by:

$$\begin{aligned} \text{Minimize} \quad & -2y_1 + y_2 + 2z \\ \text{subject to} \quad & -y_1 + y_2 - 4z \leq 0 \\ & 2y_1 + y_2 - 14z \leq 0 \\ & y_2 - 6z \leq 0 \\ & y_1 + 3y_2 + 4z = 1 \\ & y_1, y_2, z \geq 0 \end{aligned}$$

The reader can verify that $y_1 = \frac{7}{11}$, $y_2 = 0$, and $z = \frac{1}{11}$ is an optimal solution to the above linear program. Hence, the optimal solution to the original problem is $x_1 = y_1/z_1 = 7$ and $x_2 = y_2/z_2 = 0$.

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3 4
N
2 3

(3)

max $\frac{(3x_1+5x_2-3x_3)}{(x_1-4x_2+2x_3+3)}$

s.t.

$2x_1+x_2+6x_3 \leq 3$

$-5x_1+4x_2-x_3 \geq 5$

$x_1, x_2, x_3 \geq 0$

sol. $(x_1 = \frac{4}{13}, x_2 = \frac{25}{13}, x \geq 0$

obj. $f = \frac{-93}{27}$

$\frac{21}{13} + \frac{125}{13}$

$\frac{4}{13} - \frac{100}{13} + 3$

$= \frac{146}{13}$

$= \frac{146}{54} = \frac{43}{27}$

$-\frac{93}{13} + 3$

$\frac{93}{39}$

$54$

$\frac{54}{13}$ 8 $91$

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1) $\max \; \frac{x_1 + 2x_2 + 4}{x_1 + x_2 + 2}$

s.t.

$2x_1 + x_2 \le 2$

$x_1, x_2 \ge 0$

solve by Fractional programming method.

sol. ( $x_1 = 0, x_2 = 2$ ) $z_{obj} = 2$

( $x_1 + x_2 + 2 > 0, \forall x \in S$ )

2) $\max \; \frac{(x_1 + 2x_2)}{(4x_1 + 3x_2 + 3)}$

s.t.

$x_1 + x_2 \le 2$

$-x_1 + x_2 \le 1$

$x_1, x_2 \ge 0$

$(\frac{1}{2}, \frac{3}{2})$ , $W = \frac{4}{19}$

$4x_1 + 3x_2 + 3 > 0 \quad \forall x \in S$

sol. ( $x_1 = \frac{1}{2}, x_2 = \frac{3}{2}$ ) $W = \frac{4}{19}$ )

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$Ax = b$
$x = A^{-1}b$

Write the dual for the following primal (LP) problem

$\max Z = -5x_1 + 2x_2$

i) s.t. $-x_1 + x_2 \leq -2$

$2x_1 + 3x_2 \leq 5$

$x_1, x_2 \geq 0$

ii) $\min Z = 6x_1 + 3x_2$

s.t. $6x_1 - 3x_2 + x_3 \geq 2$

$3x_1 + 4x_2 + x_3 \geq 5$

$x_1, x_2, x_3 \geq 0$

iii) $\max Z = x_1 + x_2$

s.t. $2x_1 + x_2 = 5$

$3x_1 - x_2 = 6$

$x_1, x_2$ free

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Wednesday 8 أغسطس ٨ الأربعاء

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The operations Research problem solver

staff of Research and Education Association,

Dr. M. Fogiel, Director

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